

Robot Pencil Grasping and Image Drawing via Jacobian-Based Control

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- Drawing is a precision contact task — directly benchmarks control accuracy
- Extends classic manipulation (pick-and-place) to a creative, verifiable output
- One unified pipeline spanning perception through motion control

Perception

Grasp

Trajectory

Control

Unified Pipeline

Can a standard 6-DoF arm grasp a pencil and reproduce a bitmap image on paper using only Jacobian-based control?

Three Success Criteria

- | | |
|---------------------------|---------------------------------------|
| 1. Grasp Success Rate | $\text{\#success} / N \text{ trials}$ |
| 2. Pen-Tip Tracking Error | RMS and peak (mm) |
| 3. Drawing Similarity | Directed Hausdorff distance (mm) |

Frames & Forward Kinematics

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SE(3) Transform

$$T = \begin{bmatrix} R & p \\ 0 & 1 \end{bmatrix}, \quad R \in SO(3), \quad p \in \mathbb{R}^3$$

Forward Kinematics

$$T_{ee}(q) = \text{product of } T_i(q_i) \text{ for } i = 1..n$$

Differential Kinematics

$$\dot{x} = \begin{bmatrix} \omega \\ v \end{bmatrix} = J(q) \dot{q}$$

Jacobian Pseudoinverse Control

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Reference Twist (error feedback)

$$\mathbf{x}_{i_ref} = [k_{\omega} * e_{\omega} ; k_v * e_v]$$

Joint Velocity Command

$$\mathbf{q_dot} = \mathbf{J}^+ \mathbf{x}_{i_ref} + (\mathbf{I} - \mathbf{J}^+ \mathbf{J}) \mathbf{q_dot_ref}$$

Pseudoinverse

$$\mathbf{J}^+ = \mathbf{J}^T (\mathbf{J} \mathbf{J}^T)^{-1} \quad (\text{Moore-Penrose})$$

Joint Integration

$$\mathbf{q} \leftarrow \text{clip}(\mathbf{q} + \mathbf{q_dot} * dt, \mathbf{q_min}, \mathbf{q_max})$$

Control gains: $k_v = 5 \text{ s}^{-1}$, $k_{\omega} = 2 \text{ s}^{-1}$, $dt = 10 \text{ ms}$

Drawing Trajectory Generation

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Step 1

Canny edge detection on input bitmap I

Step 2

Trace edges into K polylines $S = \{\text{polyline}_k\}$

Step 3

Lift 2-D point to 3-D in paper frame ($z = z_{\text{down}}$ or $z = z_{\text{up}}$)

Step 4

Transform to world frame via T_{paper}

Algorithm: Drawing Controller

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Input: q_0 , T_{paper} , stroke set S *Output:* joint trajectory $q(t)$

```
q <- q_0
for each stroke polyline in S:
  move pen to start (z = z_up)
  lower pen          (z = z_down)
  for each waypoint p in polyline:
    compute desired tip pose  $T^*_{tip}$ 
    compute error e and reference twist  $\xi_{ref}$ 
    q_dot <-  $J(q) \xi_{ref} + (I - J^+ J) \dot{q}_{ref}$ 
    q <- clip( $q + q\_dot * dt$ ,  $q_{min}$ ,  $q_{max}$ )
  raise pen          (z = z_up)
```

Software Architecture (ROS 2)

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perception_node

Publishes T_{pencil} , T_{paper}
(depth camera or sim ground truth)

planner_node

Receives poses + bitmap I
Publishes reference twist array

controller_node

Subscribes to ref. twists
Runs pseudoinverse @ 100 Hz
Publishes JointTrajectory

Experiment 1: Grasp Success Rate

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Setup

- PyBullet simulation, UR5 6-DoF arm
- 50 randomised pencil poses (10 by 10 cm region)
- Top-down antipodal grasp strategy

Condition	Successes	Rate
Nominal (known pose)	48 / 50	96%
+/- 5 mm pose noise	42 / 50	84%
+/- 10 mm pose noise	34 / 50	68%

Takeaway: 96% under known pose; degrades predictably with perception noise.

Experiment 2: Trajectory Tracking Error

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Control: $k_v = 5 \text{ s}^{-1}$, $k_{\omega} = 2 \text{ s}^{-1}$, $dt = 10 \text{ ms}$

Shape	RMS Error (mm)	Peak Error (mm)
Line (10 cm)	1.3	3.1
Square (10 by 10 cm)	2.8	7.4 -> 4.2*
Circle (r = 5 cm)	2.1	5.9

* Corner-smoothing spline reduces square peak error from 7.4 -> 4.2 mm

Observation

Corner discontinuities drive peak error in the square case. Applying a corner-smoothing spline brings peak error within the sub-5 mm threshold for all shapes.

Experiment 3: Drawing Quality

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- Target image: house silhouette (2-stroke line art)
- Metric: directed Hausdorff distance between target edges and recorded pen path

3.7 mm

Mean Directed Hausdorff Distance

8.2 mm

95th-Percentile Distance

Nearly all produced strokes fall within one stroke-width of the target — visually faithful reproduction.

Nullspace Projection Benefit

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Comparison: Controller With vs. Without Nullspace Term

Condition	Joint Saturation Rate
Without nullspace term	52%
With nullspace term	6%

Tracking error: no measurable increase with nullspace term enabled.

Key Insight: Nullspace projection is a 'free' improvement — reduces joint saturation from 52% to 6% at zero cost to tracking accuracy.

Current Failure Modes

Singularity proximity

Damped pseudoinverse $J^+_{\lambda} = J^T (J J^T + \lambda I)^{-1}$

Integration drift

Periodic re-zeroing against joint-space IK solution

Grasp degradation

Depth-camera pose estimation (e.g., PoseCNN)

Future Work

- Hardware deployment on physical UR5
- Learned stroke primitives for curved / calligraphic marks
- Force/torque sensing for contact pressure regulation

Complete Pipeline

Perception, grasp, trajectory, and control built end-to-end

Sub-cm Tracking

Jacobian pseudoinverse achieves $\text{RMS} < 3 \text{ mm}$ on all test shapes

Nullspace Benefit

Joint saturation: 52% to 6% with zero tracking cost

Grasp Reliability

96% under known pose; degrades predictably with noise

YES — a Jacobian-based pipeline can reliably grasp a pencil and reproduce images with low tracking error.

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